Statistical Mechanical Considerations on Optical and Thermal Processes

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Abstract

Optics and thermodynamics have long been regarded as independent research domains; however, their deeper interconnections under the framework of statistical mechanics has not been fully clarified. This paper proposes that, from the perspective of statistical mechanics, the essence of the system's temperature increase lies in the growth of the average number of quanta within phase cells. Moreover, absorption of optical radiation and heat transfer are two equivalent ways of raising the system's temperature. Both mechanisms achieve the temperature increase by increasing the average number of quanta in the system's phase cells, representing two different manifestations of the same quantum statistical mechanism. Finally, two concrete examples are provided to further illustrate the statistical equivalence of these two mechanisms.

1 Introduction

Since the establishment of thermodynamics and statistical mechanics, heat has generally been regarded as a mode of energy transfer, while optics primarily focuses on electromagnetic radiation and the quantization of energy. Following Planck's formulation of the blackbody radiation law [1], Einstein was inspired to provide a correct explanation of the heat capacity of solids [2], linking optical phenomena with thermal phenomena. Nevertheless, within the framework of statistical mechanics, the relationship between optical and thermal phenomena remains insufficiently clarified.

It is commonly considered that absorption of optical radiation and heat transfer are entirely distinct operational mechanisms; the former involves interactions between photons and matter, while the latter involves energy exchange between systems. From the perspective of statistical mechanics, the system's temperature depends on the distribution of energy among the phase cells and is independent of the source of external energy. Whether the external energy comes from absorption of optical radiation or from heat transfer, the microscopic behavior of the system manifests as a redistribution of energy within phase cells, indicating that both processes leading to temperature increase are statistically equivalent.

Therefore, from the perspective of statistical mechanics, the essence of a temperature increase lies in the growth of the average number of quanta within phase cells, and both absorption of optical radiation and heat transfer constitute statistically equivalent mechanisms for achieving this increase. The following sections will further elaborate on this viewpoint.

2 Theoretical Framework and Statistical Derivation

Phase space is divided into phase cells of size h^3. Bose proposed that the microscopic distribution on these phase cells determines the macroscopic properties of a radiation system [3].

In a radiation system, there are quanta of different frequencies, whose energy is E^s, where s ranges from 0 to ∞. If the volume of a phase cell is denoted by $dxdydzdp_xdp_ydp_z$, and its size is h^3, then, once the spatial range and momentum range are given, the number of phase cells in the system is [4]

$$\Gamma_{\rm tot} = \frac{1}{h^3}\int dxdydzdp_xdp_ydp_z. \tag{1}$$

Now suppose that, within a unit energy interval, the number of phase cells is Δs, and that Δn quanta are distributed over them, where both Δs and Δn are very large numbers. We use $p_r\Delta s$ to denote the number of phase cells in Δs in which r quanta appear. p_r also denotes the probability that r quanta appear in the s-th phase cell; therefore, we use p_r^s to replace p_r. Clearly, for all phase cells s, the following condition is satisfied,

$$\sum_r p_r^s = 1. \tag{2}$$

Given p_r^s, the number of possible distributions of Δn among Δs phase cells is

$$\frac{\Delta s!}{\prod_{r=0}^{r=\infty}(p_r^s\Delta s)!}. \tag{3}$$

Here the range of r is from 0 to infinity. Taking the logarithm of the above expression and then using Stirling's formula together with Eq.(2), we obtain

$$-\Delta s\sum_r p_r^s \ln p_r^s. \tag{4}$$

Therefore, for the entire radiation system, this expression may be written as a summation over all phase cells. Combining this with Boltzmann principle, one obtains the expression for the entropy S of the radiation system,

$$S = -k_{\rm B}\sum_{s,r} p_r^s \ln p_r^s, \tag{5}$$

where $k_{\rm B}$ is Boltzmann constant.

When the radiation system reaches thermodynamic equilibrium, the entropy is required to attain its maximum value, while the total energy of the system is given by

$$\overline{E} = \sum_{s,r} E^s r p_r^s. \tag{6}$$

Here, $\sum_{s,r} rp_r^s$ is the average number of quanta in the s-th phase cell. We now seek the maximum value of Eq.(5) under the constraint of Eq.(6). Introducing the Lagrange multiplier β, and taking Eq.(2) into account, we obtain

$$p_r^s = (1-e^{-\beta E^s})e^{-\beta E^s r}. \tag{7}$$

The expression for β is then determined. Substituting Eq.(7) into Eq.(5), we obtain

$$S = -k_{\rm B}\sum_s \ln(1-e^{-\beta E^s}) + k_{\rm B}\beta\overline{E}. \tag{8}$$

Combining this with the thermodynamic relation $\frac{\partial S}{\partial \overline{E}} = \frac{1}{T}$, where T denotes the system temperature, gives $\beta = \frac{1}{k_{\mathrm{B}}T}$. Therefore, combining Eq.(6) and Eq.(7), the average energy $\overline{E^s}$ in the s-th phase cell is

$$\overline{E^s} = \frac{E^s}{e^{E^s/k_{\mathrm{B}}T} - 1}. \tag{9}$$

Replacing E^s by $h\nu^s$, we obtain the expression for the average energy of a Planck oscillator.

3 Proposed Theoretical Perspective

Below, consider a system in thermal equilibrium. The system temperature is related to the average number of quanta $\overline{n^s}$ corresponding to each phase cell. When the temperature of the system increases, $\overline{n^s}$ changes. For every increase of 1K, the average number of quanta corresponding to each phase cell increases by

$$\Delta\overline{n^s} = \frac{e^{\frac{E^s}{k_{\mathrm{B}}T}}}{(e^{\frac{E^s}{k_{\mathrm{B}}T}} - 1)^2} \frac{E^s}{k_{\mathrm{B}}T^2}. \tag{10}$$

Eq.(10) indicates the essence of temperature increase: an increase in the average number of quanta within phase cells. In addition, from a macroscopic perspective, absorption of optical radiation and heat transfer are two main ways of increasing the temperature of a system. However, from the perspective of statistical mechanics, these two ways of heating the system are equivalent: both increase the system temperature by increasing the average number of quanta within phase cells.

Therefore, in statistical mechanics, there is no essential distinction between these two ways of increasing temperature, which involve optical processes and thermal processes, respectively. They are two forms of manifestation under the same quantum statistical mechanism.

4 Examples and Discussion

4.1 The Statistical Equivalence of the Two Ways in Vibrational Heat Capacity

To further clarify the above theoretical viewpoint, the following takes the vibrational heat capacity of a diatomic molecule as an example to explain, within the framework of statistical mechanics, the statistical equivalence involving optical and thermal processes.

A diatomic molecule may be regarded as a one-dimensional linear harmonic oscillator formed by the relative vibration of two atoms. The average number of quanta contained in the i-th oscillator is expressed as

$$\overline{n_i} = \frac{1}{e^{\frac{\Theta_{\mathrm{v}}}{T}} - 1}. \tag{11}$$

Here, $\Theta_{\mathrm{v}} = E/k_{\mathrm{B}}$, which is called the characteristic temperature of molecular vibration (it is assumed that the characteristic temperature is the same for any oscillator, and the

energy carried by a oscillator is written as E). Taking the derivative with respect to temperature gives

$$\frac{\partial \overline{n_i}}{\partial T} = \frac{e^{\frac{\Theta_{\mathrm{v}}}{T}}}{(e^{\frac{\Theta_{\mathrm{v}}}{T}}-1)^2}\frac{\Theta_{\mathrm{v}}}{T^2}. \tag{12}$$

It can be seen that, when the system temperature is increased by 1K, the average number of quanta in each oscillator increases by the amount given in Eq.(12). This indicates the reason for the temperature increase of a diatomic molecular system: the increase in the average number of quanta within the phase cells of the system. It should also noted here that the increase in the average energy within the phase cells of the system comes from the energy transferred to the system by the external environment. From the perspective of statistical mechanics, whether this occurs through absorption of optical radiation or through heat transfer, the system's response to such external energy is the same in a statistical sense.

In addition, multiplying Eq.(12) first by E and then by N_{A} gives the increase in the vibrational energy of 1 mole of diatomic molecules. This increase is also the energy that the external environment needs to transfer to the system; and it is also similar to the molar vibrational heat capacity c_V^{v} given by Mr. Einstein [2]:

$$c_V^{\mathrm{v}} = \mathrm{R}\frac{e^{\frac{\Theta_{\mathrm{v}}}{T}}}{(e^{\frac{\Theta_{\mathrm{v}}}{T}}-1)^2}(\frac{\Theta_{\mathrm{v}}}{T})^2. \tag{13}$$

where $\mathrm{R} = k_{\mathrm{B}}N_{\mathrm{A}}$.

Thus, it can be seen that, whether absorption of optical radiation or heat transfer is adopted, for the same diatomic molecular system, the energy required to raise its temperature by 1K is given by the vibrational heat capacity shown in Eq.(13). This shows the statistical equivalence of optical and thermal processes involved in vibrational heat capacity.

4.2 The Statistical Equivalence of the Two Ways in Other Systems

First, consider the statistical equivalence of the two ways of increasing the temperature of a blackbody system. If energy enters the blackbody through heat transfer, this energy will be absorbed by the blackbody and redistributed among the phase cells. This way of increasing the system temperature is not essentially different from absorption of optical radiation; both increase the system temperature by increasing the number of quanta within the phase cells of the system.

Second, the two ways of increasing the temperature of metals or solids also have statistical equivalence. Similarly, within the framework of statistical mechanics, absorption of optical radiation and heat transfer both involve, at the microscopic level, the redistribution of the system's total energy and an increase in the number of quanta in its phase cells. Therefore, the two ways are statistically equivalent in raising the temperature of metals or solids.

References

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